INCH-POUND
MIL-DTL-24231/18E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/18D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, PLUG, TYPE I, MOLDED, 65-CONDUCTOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 5.

2. The sealing plug assembly is designed to seal off the plug against hydrostatic pressure as high as 2,000 lbf/in^2 and shall remain in place until assembly of the plug with the receptacle.

3. Connections of wires in cable TSP-31 to sockets shall be made in accordance with the applicable table in the appendix of MIL-DTL-24231.

4. All dimensions are in inches.

5. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½degree.

6. These connectors are for maintenance purposes only.

7. Ni-Cu plug sleeves shall be coated with plasma spray non-conductive material in accordance with S9320-AM-PRO-030/MLDG.

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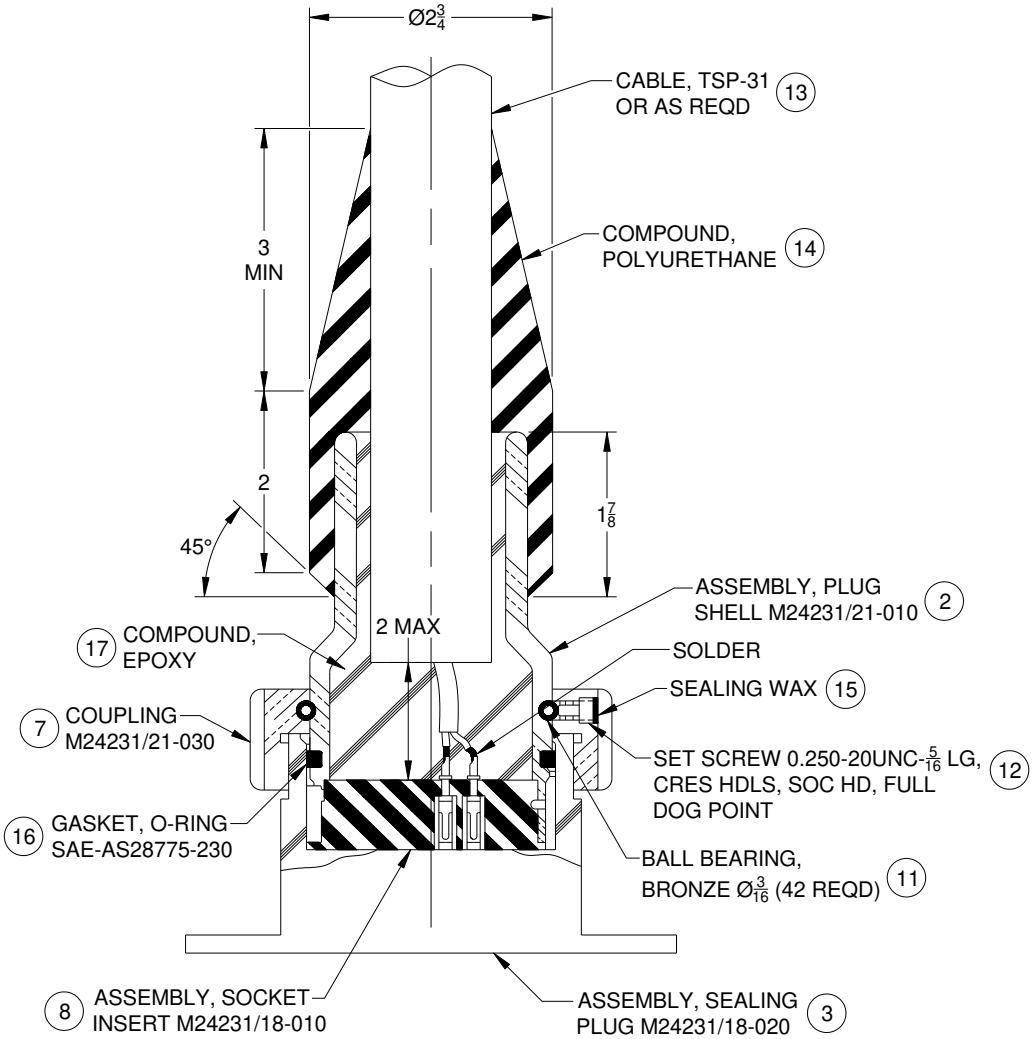


FIGURE 1. Connector, plug, 65-conductor. (1)

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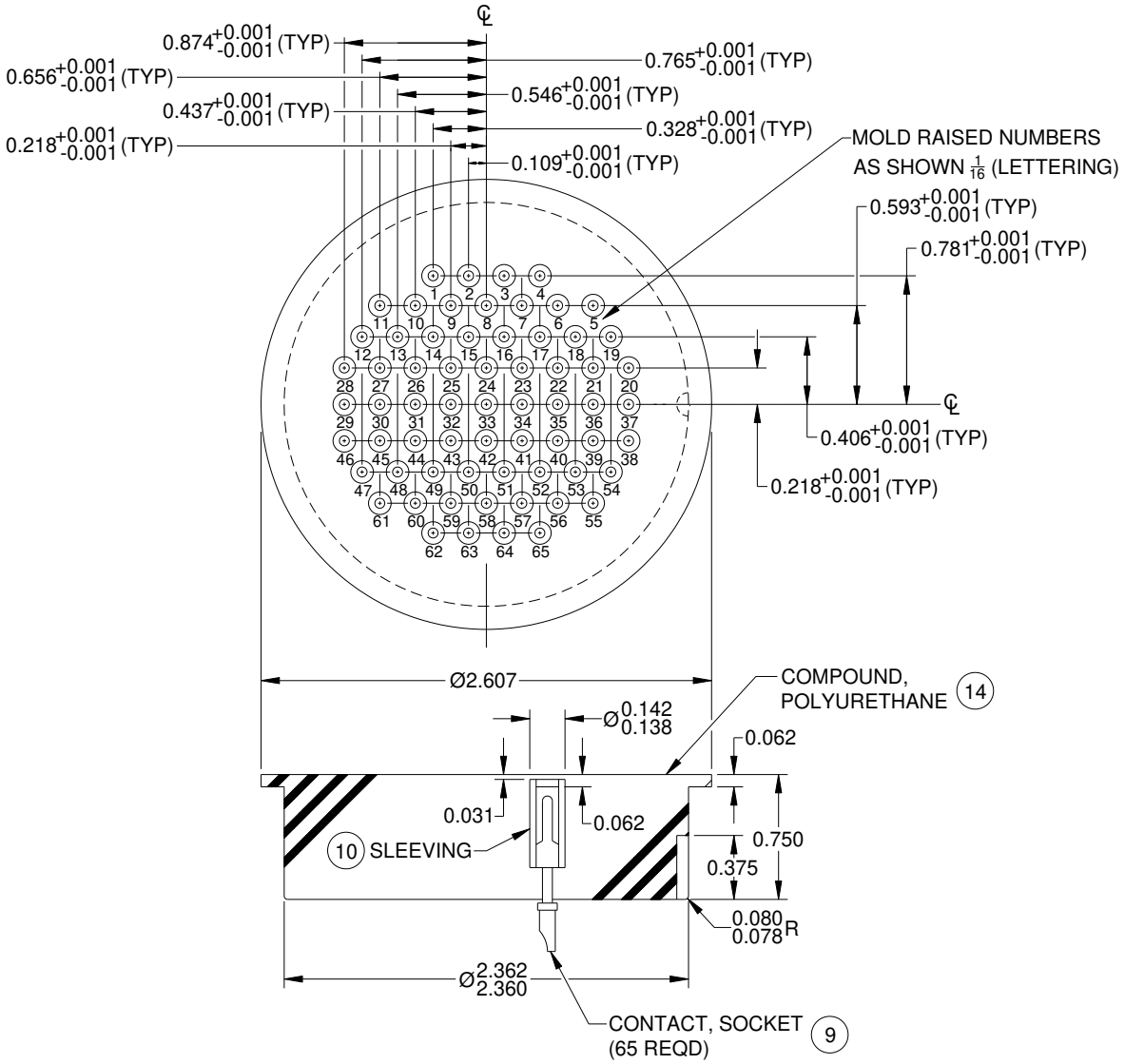


FIGURE 2. Assembly, socket insert – M24231/18-010. (8)

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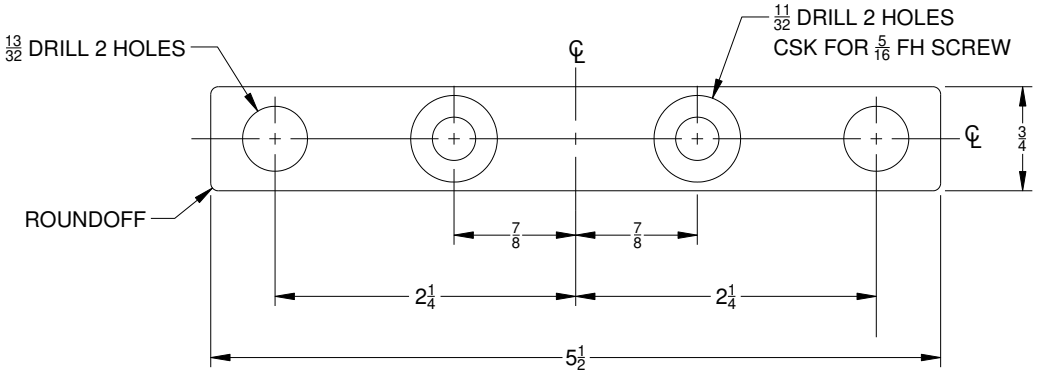


FIGURE 3. Handle, 3/16-inch thick. (4)

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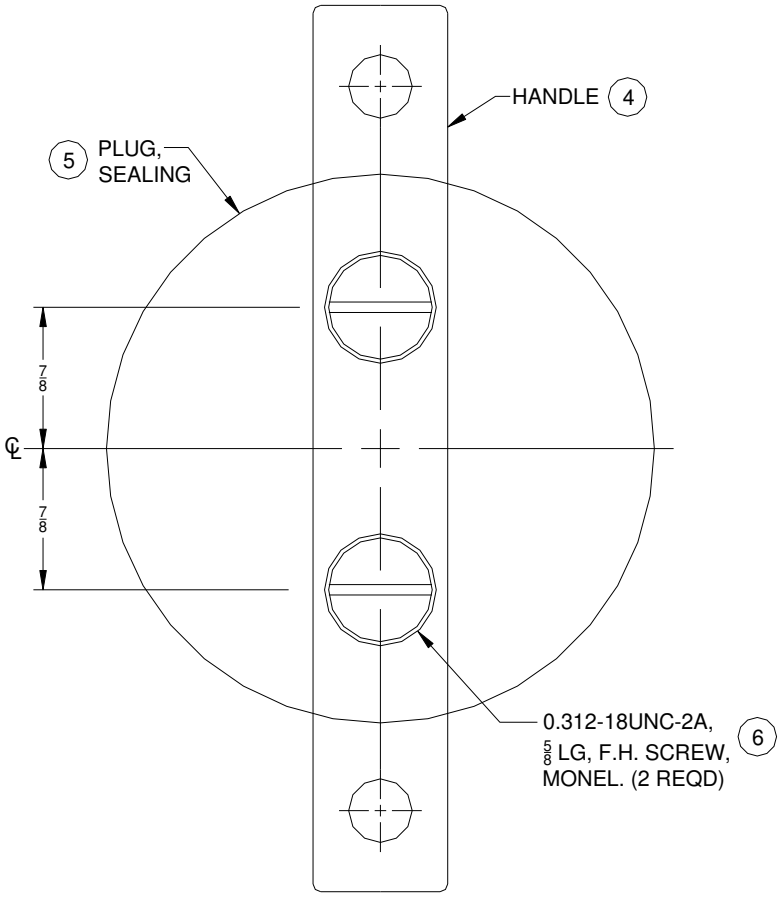


FIGURE 4. Assembly, sealing plug – M24231/18-020. 3

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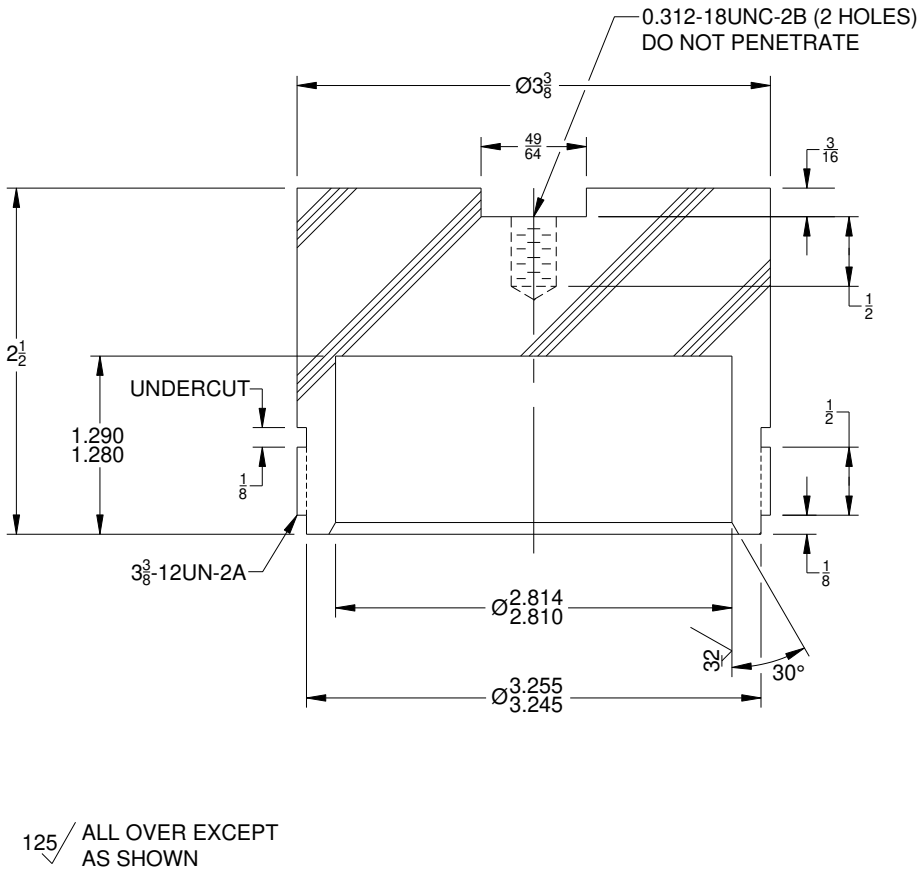


FIGURE 5. Plug, sealing. (5)

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TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/18- 3/	Material/specification	Remarks
		001		
1	Plug, 65-conductor, symbol no. 1139	1	------	See figure 1.
2	Shell, plug	1	M24231/21-010	See figure 1.
3	Sealing plug assembly	1	See items 4, 5, and 6.	See figure 4 and requirement 2.
4	Handle, 3/16 thick	1	Phosphor bronze, hard, coml	See figure 3.
5	Sealing plug	1	Type II, Grade GV of ASTM D1784	See figure 5.
6	Screw, F.H. 0.312-18UNC-2A, 5/8 long	2	Ni-Cu, Class A, QQ-N-281	See figure 4.
7	Coupling	1	M24231/21-030	See figure 1.
8	Insert, assembly, socket	1	See items 9, 10, and 14.	See figure 2.
9	Contact, socket	65	MIL-DTL-24231	See requirement 3.
10	Sleeving, socket no. 9, 0.118 id by 9/16 long	65	Type F, Form U_a, Grade a, Class II, Category 1, color clear, MIL-I-631	See figure 2.
11	Ball bearing, 3/16 inch diameter	42	Bronze, ASTM F2215	See figure 1.
12	Screw, set, 1/4-20UNC by 5/16 Lg handles, soc-hd, full dog point	1	CRES, coml	See figure 1.
13	Cable, TSP-31	As required	MIL-DTL-915/22	See figure 1.
14	Compound, polyurethane	As required	4/	See figure 1.
15	Wax, sealing	As required	Coml	See figure 1.
16	Gasket, O-ring	1	SAE-AS28775-230 5/	Lubricate with silicone compound, SAE-AS8660; see figure 1.
17	Compound, epoxy	As required	Cast epoxy-resin compound 6/	Base EE4183 or RE2039, hardener HD-3561; see figure 1.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Molding compound to be procured in accordance with S9320-AM-PRO-020.

5/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

6/ Hysol base EE4183 or RE2039, hardener HD-3561, Hysol Division, Dexter Corporation, Olean, NY, CAGE 04347, or equal.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-067)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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